# Seokyoung Yoon

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## Education

- **Pohang University of Science and Technology (POSTECH)** *Feb 2026 - Present*
  *Mueunjae School of Undergraduate Studies* Pohang-si, Republic of Korea
  - GPA: 3.95/4.30
  - Jigok Scholarship (full tuition), maintained through academic performance
- **Jeonbuk Science High School** *Mar 2024 - Feb 2026*
  *Secondary Education* Iksan-si, Republic of Korea

## Research Experience

- **Research Intern** *Jun 2026 - Aug 2026*
  *HOLI Lab, Seoul National University*
  *Advisor: Prof. Yohan Jo* Seoul, Republic of Korea
  - Tested whether case morphology and positional encoding are substitutable channels for grammatical role.
  - Ablated positional embeddings, masked attention at word boundaries, and patched activations across layers in BERT.
  - Wrote the work up as an extended abstract and submitted it to BlackBoxNLP.
  - Learned to audit instruments before trusting results, invalidating a released checkpoint.
- **R&E Team Leader** *Mar 2024 - Dec 2024*
  *InfoSec Team, Jeonbuk Science High School* Iksan-si, Republic of Korea
  - Thesis: *A Research on Deauthentication Attack Detection and Notification*
  - Directed a team of four from initial hypothesis design to final experimental validation.
  - Assigned tasks based on team members' strengths and mediated technical discussions to reach project milestones.

## Honors& Awards

- **Gold Prize, Hackathon Category, The 7th Korea Code Fair** *Dec 2025*
  *Minister's Award, Ministry of Science and ICT*
  - Developed an emoji-based emotion tracking application supporting adolescent mental health.
  - Led a team of three from UI/UX design to final system integration.
  - Designed the AI architecture around a custom exponential moving average, validated through a proof of concept.
- **Honorable Mention, The 71st National Science Exhibition** *Nov 2025*
  *National Science Museum*
  - Researched document-based PDF adversarial attacks designed to mitigate AI overdependence.
  - Built a web service implementing the proposed methods to validate practical feasibility.
  - Led the team, managing core methodology design and system architecture.

## Teaching& Mentoring

- **Physics Mentor** *Apr 2026 - Jul 2026*
  *University Student Mentoring Program (Science D), Pohang High School* Pohang-si, Republic of Korea
  - Taught high school physics over 20 hours of sessions.
  - Provided academic guidance and support for coursework and school life.

## Skills

- **Machine Learning:** PyTorch, NumPy, SciPy, Pandas, Matplotlib, PySpark
- **Programming Languages:** Python, C, C++, Rust, JavaScript, SQL, Elixir
- **Software Engineering:** React.js, Next.js, Svelte, FastAPI, Express.js, PostgreSQL, Redis, Docker, Git, Google Cloud
- **Research Tools:** LaTeX, Typst, Zotero, SymPy, Markdown
- **Laboratory Techniques:** GC-MS, HPLC, NMR, XRF, Gel Electrophoresis
- **Fabrication:** CAD Design, Laser Cutting, 3D Printing, Lathe

## Languages

**Languages:** Korean (Native), English (Proficient), Spanish (Basic)